

SPRAGUE, MB, Canada

WMO: 714490

Lat: 49.0236N

Lon: 95.5983W

Elev: 1079

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-22.5	-35.1	0.8	-28.1	-28.9	1.1	-22.2	23.0	7.6	20.7	6.3	3.9	230	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	84.8	70.3	81.8	67.9	79.1	66.4	73.2	81.1	70.9	78.6	68.8	76.1	7.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	117.0	78.4	68.2	107.9	75.5	66.0	99.8	73.2	37.6	81.2	35.5	78.6	33.6	76.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.0	15.9	-36.5	90.2	7.1	2.4	-41.6	91.9	-45.8	93.4	-49.8	94.7	-54.9	96.5		
			-37.1	76.5	7.2	2.5	-42.2	78.3	-46.4	79.8	-50.5	81.2	-55.7	83.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	3.9	8.1	22.8	38.3	50.9	61.3	66.0	63.8	55.2	41.9	25.8	10.7
	DBStd	24.22	15.40	14.25	13.95	10.23	8.55	6.59	5.55	6.01	8.36	8.84	11.94	13.96
	HDD50	6174	1430	1172	845	368	92	4	0	0	40	280	726	1217
	HDD65	10216	1895	1592	1308	803	442	145	56	94	308	716	1175	1682
	CDD50	1637	0	0	2	16	122	344	496	428	197	31	1	0
	CDD65	200	0	0	0	0	6	34	87	57	15	1	0	0
	CDH74	2161	0	0	0	5	103	383	873	606	181	10	0	0
	CDH80	456	0	0	0	0	17	74	198	130	36	1	0	0
	WSAvg	6.8	6.9	7.1	7.3	8.0	7.7	6.4	5.7	5.5	6.2	7.3	7.3	6.8
	PrecAvg	24.5	0.9	0.7	1.1	1.6	2.6	3.9	3.8	3.3	2.6	1.7	1.1	0.9
(15) Precipitation	PrecMax	31.4	3.3	2.2	3.0	4.2	6.0	8.3	7.7	10.0	5.5	5.1	3.9	2.4
	PrecMin	14.2	0.1	0.1	0.1	0.0	0.5	0.2	1.5	0.6	0.3	0.3	0.0	0.2
	PrecStd	4.4	0.7	0.6	0.7	1.1	1.5	2.1	1.5	2.4	1.7	1.1	0.9	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	42.3	61.7	73.1	82.3	86.7	88.7	88.4	84.7	74.1	58.4	39.6
		MCWB	35.6	37.4	50.7	54.5	63.3	72.2	72.6	72.6	69.3	59.4	49.0	35.3
	2%	DB	33.4	37.3	50.4	66.3	76.6	81.7	84.9	83.4	79.1	67.5	50.4	35.4
		MCWB	31.5	34.2	43.3	50.6	59.2	67.2	70.9	69.2	65.7	56.5	44.0	33.0
	5%	DB	29.8	33.6	45.0	60.9	71.9	78.3	81.9	80.3	74.8	61.6	45.9	32.6
		MCWB	28.1	31.5	39.4	47.9	56.9	65.0	69.1	67.9	63.0	52.6	40.9	31.1
	10%	DB	25.5	29.8	40.2	56.0	67.5	75.3	79.0	77.2	70.4	56.2	41.2	29.5
		MCWB	24.3	27.8	36.0	44.9	54.8	63.6	67.5	65.7	61.1	49.5	37.7	28.1
	0.4%	WB	35.6	38.3	51.7	55.8	66.6	75.5	76.2	75.3	72.0	62.2	50.1	36.0
		MCDB	39.0	40.9	57.7	69.2	76.1	83.2	84.2	83.7	80.5	68.0	55.4	38.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	34.6	44.0	52.1	62.6	70.9	73.6	72.0	68.3	57.5	45.3	33.3
		MCDB	33.2	36.9	49.3	63.7	71.6	78.4	81.2	80.2	75.0	66.3	49.5	34.9
	5%	WB	28.2	31.7	39.5	48.9	59.4	67.3	71.3	69.8	64.7	53.4	41.2	31.1
		MCDB	29.8	33.6	44.4	58.6	67.2	74.3	78.8	77.4	71.2	59.9	45.3	32.7
	10%	WB	24.3	28.0	36.2	45.9	56.7	64.9	69.2	67.7	62.1	49.6	37.6	28.2
		MCDB	25.5	29.7	39.7	55.7	65.2	72.2	76.7	74.5	69.1	55.6	41.0	29.5
	5% DB	MDBR	19.2	21.3	20.4	21.1	22.0	20.5	21.8	22.5	21.6	17.5	15.0	16.3
		MCDBR	17.9	20.0	21.8	30.0	29.2	25.8	25.2	26.2	27.9	26.6	20.9	17.0
		MCWBR	16.1	16.7	15.0	17.6	15.1	13.8	13.6	14.3	15.8	16.2	14.6	14.6
		MCWBR	17.3	18.4	21.0	26.9	22.9	21.1	22.1	22.9	23.2	23.3	18.8	16.1
(35) Mean Daily Temperature Range	5% WB	MCWBR	15.7	15.4	14.9	16.7	13.9	13.2	13.7	14.0	14.5	15.5	13.7	14.1
	taub		0.262	0.295	0.293	0.343	0.359	0.376	0.380	0.374	0.346	0.303	0.275	0.256
		taud	2.289	2.193	2.404	2.381	2.400	2.379	2.386	2.411	2.479	2.565	2.491	2.362
(42) Clear-Sky Solar Irradiance	Ebn at Noon		254	268	294	287	286	280	277	274	272	270	253	243
		Edh at Noon	22	31	30	35	36	37	36	34	29	22	18	18
	RadAvg		435	814	1194	1520	1708	1797	1918	1594	1176	690	384	309
(45) All-Sky Solar Radiation	RadStd		47	64	117	139	114	142	94	116	95	99	47	34

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	-1.62	N/A	N/A	+119	N/A
Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page